

Reg. No.

B.Tech. DEGREE EXAMINATION, MAY 2023

First Semester

21BMC101J – BIOMEDICAL SENSORS*(For the candidates admitted from the academic year 2022-2023)***Note:**

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part – B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)Answer **ALL** Questions

	Marks	BL	CO	PO
1. The block that makes physical contact with physical quantity to be measured is (A) Primary sensing element (B) Variable conversion element (C) Variable manipulation element (D) Data transmission element	1	1	1	1,2
2. Which of the following is a measuring instrument? (A) Ammeter (B) Voltmeter (C) Attenuator (D) Ammeter and voltmeter	1	1	1	1,2
3. Resistance can be measured using (A) Ammeter (B) Ohmmeter (C) Wattmeter (D) Tachometer	1	1	1	1,2
4. Which category of instruments are best suited for engineering applications? (A) Analog instruments (B) Physical instruments (C) Digital instruments (D) Chemical instruments	1	1	1	1,2
5. Which converts one form of energy into another form? (A) Transistor (B) Transformer (C) Transducer (D) Transmitter	1	1	2	1,4
6. Which of the following is not a temperature transducer? (A) RTD (B) Thermistor (C) Load cell (D) Thermocouple	1	1	2	1,4
7. Seeback effect is associated with (A) RTD (B) Load cell (C) Strain gauge (D) Thermocouple	1	1	2	1,4
8. Which possess negative temperature coefficient of resistance? (A) RTD (B) Thermistor (C) Thermocouple (D) IR thermography	1	1	2	1,4

9. The movable part of LVDT is	1	1	4	1,5
(A) Core				
(B) Primary winding				
(C) Secondary winding				
(D) Isolator				
10. The output of piezoelectric transducer is an form of	1	2	3	1,2
(A) Current				
(B) Voltage				
(C) Resistance				
(D) Capacitance				
11. The capacitance of a capacitor is inversely proportional to	1	2	3	1,2
(A) Permittivity				
(B) Area of cross section				
(C) Distance between the plates				
(D) Permittivity of vaccum				
12. The output of strain gauge is in form of	1	2	3	1,2
(A) Voltage				
(B) Current				
(C) Power				
(D) Resistance				
13. Which is an optoelectronic transducer?	1	2	5	1,2
(A) LDR				
(B) LVDT				
(C) Strain gauge				
(D) Load cell				
14. Photovoltaic effect is associated with	1	2	5	1,2
(A) LDR				
(B) Phototransistor				
(C) Opto isolator				
(D) Solar cell				
15. Light is made to fall on a phototransistor on	1	2	5	1,2
(A) Base				
(B) Emitter				
(C) Collector				
(D) Bias voltage				
16. Photo diode operates on the principle of	1	2	5	1,2
(A) Photo electric effect				
(B) Piezo electric effect				
(C) Total internal reflection				
(D) Snell's law				
17. Oximeter measures	1	1	6	1,4
(A) Oxygen saturation in blood				
(B) Oxygen dimensions in blood				
(C) Oxygen pH in blood				
(D) Oxygen excitation in blood				
18. Which is a non-contact temperature sensor?	1	1	6	1,4
(A) RTD				
(B) Thermistor				
(C) Thermocouple				
(D) Infrared sensor				
19. Heart rate of a healthy adult ranges about	1	1	6	1,4
(A) 75 bpm				
(B) 90 bpm				
(C) 60 bpm				
(D) 92 bpm				
20. The BP of healthy adult ranges from	1	1	6	1,4
(A) 120/80 mm Hg				
(B) 80/120 mm Hg				
(C) 140/120 mm Hg				
(D) 120/140 mm Hg				

PART – B (5 × 8 = 40 Marks)

Answer ALL Questions

	Marks	BL	CO	PO
21. a. Classify instruments based on various factors and explain them.	8	4	1	1,2
(OR)				
b. Summarize the salient features of medical instruments.	8	2	1	1,2
22. a. Explain the construction and operating principle of RTD.	8	4	2	1,4
(OR)				
b. Demonstrate temperature measurement using thermocouple.	8	2	2	1,4
23. a. Describe the construction and operating principle of LVDT.	8	4	4	1,5
(OR)				
b. How measurement of physical quantities can be achieved by means of capacitive transducer?	8	3	3	1,3
24. a. Explain the operating principle of photovoltaic cell.	8	4	5	1,2
(OR)				
b. Illustrate the working of a phototransistor circuit.	8	3	5	1,2
25. a. Discuss the importances of pulse oximetry in health assessment.	8	2	6	1,4
(OR)				
b. Write the procedure for measurement of blood pressure of an individual.	8	2	6	1,4

PART – C (1 × 15 = 15 Marks)

Answer ANY ONE Questions

	Marks	BL	CO	PO
26. Explain the operating principle to measure force using a transducer, which gives output as an active electrical quantity.	15	5	3	1,3
27. Construct a system that could measure pulse rate of an individual.	15	6	6	1,4

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